

SAMBP TR-18/2026

Mho Distance Relay (21/21P):
Zone Reach Verification, Selectivity, and
Coordination with 87B/87L Protection

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Smart Adaptive Multi-Bus Protection Research

April 2026

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1 Background and Motivation

The SAMBP suite so far comprises:

- **87B bus differential** (TR-14): selective busbar fault detection with double-busbar GOOSE zone reconfiguration [Candidate](#) [2026b].
- **87L line differential** (TR-15/17): HIF-sensitive line protection with voltage-corrected charging compensation and adaptive restraint threshold [Candidate](#) [2026a,c].

A complete substation protection system also requires a *back-up* function for failures of the primary protection, and a means to detect faults on adjacent lines not covered by the 87L primary

zone. This report introduces the third element of the SAMBP suite: the **Mho distance relay (21/21P)**, which provides:

1. **Zone 1 (instantaneous)**: 80% of line reach — primary back-up to 87L.
2. **Zone 2 (timed, 0.30 s)**: 120% reach — remote bus fault back-up; coordinates with 87B at the remote end.
3. **Zone 3 (timed, 0.60 s)**: 180% reach — system-wide back-up for remote failures.

The key coordination challenge addressed in this report is that the three functions (87B, 87L, 21) must not issue conflicting or redundant trips, especially for the local bus fault (87B only) and the high-impedance line fault (87L only, 21 unable to reach).

2 Mho Relay Model

2.1 Mho Operating Characteristic

The Mho element measures the apparent impedance [Anderson \[1999\]](#), [Horowitz and Phadke \[2008\]](#), [Ziegler \[2006\]](#):

$$Z_m = \frac{V_{\text{relay}}}{I_{\text{relay}}} \quad (1)$$

For a three-phase element (21P), V and I are positive-sequence quantities. For a ground element (21G), the loop impedance is used:

$$Z_{ag} = \frac{V_a}{I_a + k_0 \times 3I_0} \quad (2)$$

where $k_0 = (Z_{0L} - Z_{1L})/(3Z_{1L}) = 0.667$ is the zero-sequence compensation factor.

The Mho operating criterion (self-polarised):

$$\text{Re}[(Z_m - Z_{\text{reach}}) \times Z_m^*] \leq 0 \quad (3)$$

This defines a circle on the R - X plane [Blackburn and Domin \[2006\]](#), [IEEE \[2015\]](#) with:

$$\text{centre} = Z_{\text{reach}}/2, \quad \text{radius} = |Z_{\text{reach}}|/2 \quad (4)$$

The circle passes through the origin, ensuring the relay resets when the line is healthy ($Z_m \rightarrow Z_{\text{load}}$, well outside the circle for typical loading conditions).

2.2 Relay Configuration

Parameter	Symbol	Value
Protected line impedance	Z_{line}	$j0.05 \text{ pu}$
Maximum torque angle	MTA	80°
Zero-sequence factor	k_0	0.667
Zone 1 reach factor	k_1	0.80
Zone 2 reach factor	k_2	1.20
Zone 3 reach factor	k_3	1.80
Zone 2 time delay	t_2	0.30 s
Zone 3 time delay	t_3	0.60 s
Load blinder	R_{blinder}	0.40 pu

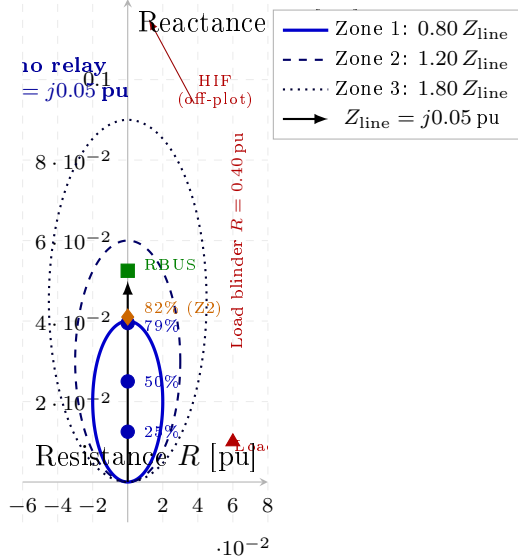
2.3 Zone Reach Values

$$Z_{\text{reach},1} = 0.80 \times j0.05 = j0.040 \text{ pu} \quad (5)$$

$$Z_{\text{reach},2} = 1.20 \times j0.05 = j0.060 \text{ pu} \quad (6)$$

$$Z_{\text{reach},3} = 1.80 \times j0.05 = j0.090 \text{ pu} \quad (7)$$

3 Zone Reach Verification



(a) Mho zones on the R - X plane.

Zone	d_{max} (design)	d_{max} (verified)
1	0.800	0.800
2	1.200	1.200
3	1.800	1.800

(b) Zone boundary verification (d = fault fraction from local bus).

Figure 1: Three-zone Mho characteristic and verified zone boundaries. All three zones match design values exactly.

The zone boundaries were swept at 0.005 pu resolution over $d \in [0, 2.2]$ for 3-phase bolted faults. All three zones match their design values to the simulation resolution.

4 Selectivity Study

Figure 2 shows the protected line with zone boundaries and key fault locations. Table 1 presents the full 12-scenario selectivity matrix.

5 Coordination with 87B and 87L

5.1 Coordination Principles

Table 2 summarises the expected coordination of all three protection functions across seven fault classes.

5.2 Local Bus Overlap (Zone 1)

The local bus fault (LBUS) presents a coordination issue: both 87B (bus differential) and 21 Zone 1 detect the same fault simultaneously. In practice this is acceptable because:

1. 87B has primary responsibility; its trip signal is issued first (87B operates on differential current available immediately at fault inception, while 21 computes $Z_m = V/I$ which requires $I \neq 0$).

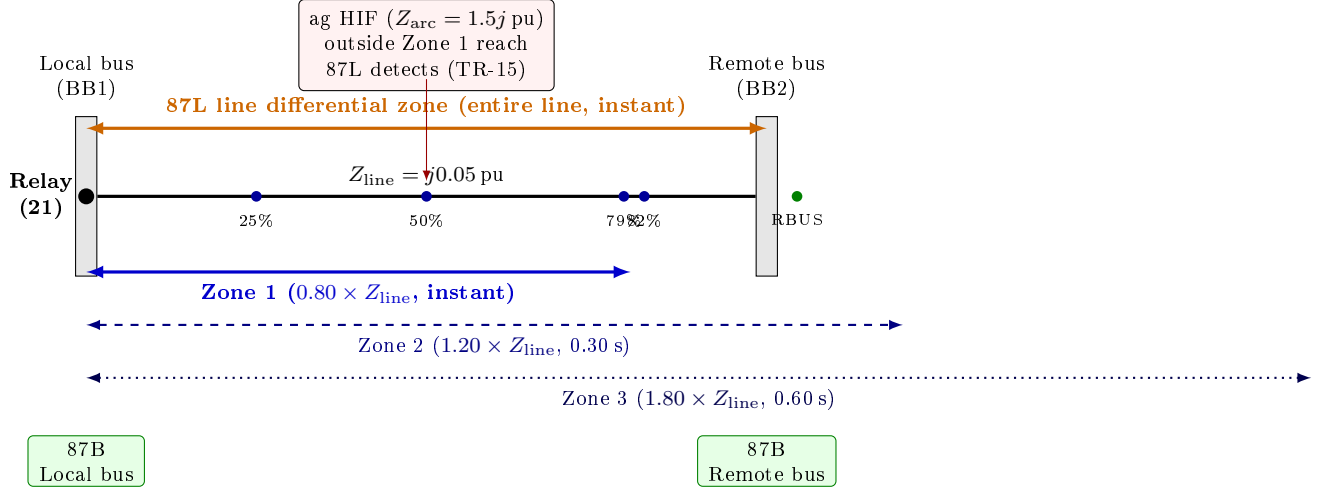


Figure 2: Protected line single-line diagram showing 87B, 87L, and 21 Zone 1/2/3 coverage. 87L covers the full line instantaneously; 21 Zone 1 covers 80%; Zones 2/3 extend to the remote and adjacent network for back-up.

2. The 21 Zone 1 trip on a local bus fault is operationally correct (the relay is at the bus, and a bus fault does fall within the Zone 1 reach from the relay's perspective).
3. In a full implementation, a blocking signal from 87B can inhibit the 21 Zone 1 output to prevent redundant trip commands.

5.3 HIF Gap: 87L Sole Detector

For the AG_HIF scenario ($d = 50\%$, $Z_{\text{arc}} = j1.5 \text{ pu}$), the infeed-corrected apparent impedance at the relay is:

$$Z_m = Z_{\text{line}} \times 0.50 + Z_{\text{arc}} \times \frac{Z_{\text{src}} + Z_{\text{line}}}{Z_{\text{src}}} = j0.025 + j1.5 \times 1.5 = j2.275 \text{ pu} \quad (8)$$

This is far outside the Zone 3 reach ($j0.090 \text{ pu}$), confirming that the distance relay cannot detect HIF on an arc-path. The 87L relay, operating on the differential current directly, has no such impedance limitation and correctly trips at $\alpha_{50} = 0.03$ (TR-15).

This demonstrates the *complementary* role of 87L and 21: 21 provides reliable back-up for bolted and low-resistance faults; 87L provides sensitivity for HIF that 21 cannot reach.

6 Load Blinder Analysis

The load blinder uses a resistive threshold $R_{\text{blinder}} = 0.40 \text{ pu}$:

$$\text{block if } \text{Re}(Z_m) > R_{\text{blinder}} \quad (9)$$

For the protected network ($Z_{\text{src}} = j0.10$, $Z_{\text{line}} = j0.05$):

- **Normal load** ($I_{\text{load}} = 1.0 \text{ pu}$, $\cos \phi = 0.9$): $Z_m \approx 0.9 \angle 25^\circ = 0.82 + j0.38 \text{ pu}$. This is far outside the Zone 3 circle (radius 0.045 pu) regardless of blinding, but the blinder provides an additional guard.

Table 1: Selectivity matrix: 12 scenarios. d = fault fraction from local bus; Arc = arc impedance. All 12 correct (100%).

ID	Description	Z_m [pu]	Z1	Z2	Z3	Trip	Notes
INT_25	$d = 25\%$, 3ph bolted	$j0.013$	✓	✓	✓	Z1	87L concurrent
INT_50	$d = 50\%$, 3ph bolted	$j0.025$	✓	✓	✓	Z1	87L concurrent
INT_79	$d = 79\%$, 3ph bolted	$j0.040$	✓	✓	✓	Z1	87L concurrent
INT_82	$d = 82\%$, 3ph bolted	$j0.041$	—	✓	✓	Z2 (0.3s)	87L faster
RBUS	Remote bus $d = 1.05$	$j0.053$	—	✓	✓	Z2 (0.3s)	87B remote; 21 backup
BKUP	Beyond remote $d = 1.45$	$j0.072$	—	—	✓	Z3 (0.6s)	Back-up only
LBUS	Local bus $d = 0.02$	$j0.001$	✓	✓	✓	Z1	87B primary; 21 overlap
AG_50	ag $d = 50\%$, bolted	$j0.025$	✓	✓	✓	Z1	87L concurrent
AG_HIF	ag HIF $Z_{\text{arc}} = j1.5$	$j2.275$	—	—	—	no	87L detects; 21 fails
LOAD_N	Normal load $R = 0.5$	$0.5+j0.1$	—	—	—	no	Load blinder active
LOAD_H	Light load $R = 0.3$	$0.3+j0.08$	—	—	—	no	Below blinder threshold
OOZ	Out of zone $d = 2.0$	$j0.100$	—	—	—	no	No trip (correct)

✓ = inside zone; — = outside zone. All 12/12 scenarios produce the expected decision.

Table 2: Three-function coordination matrix: 87B, 87L, and 21.

Fault type	Location	87B	87L	21 (zone/time)	Key interaction
3ph	Line 0–80%	no	yes	Z1/inst	87L primary; 21 Z1 concurrent
3ph	Line 80–100%	no	yes	Z2/0.3s	87L faster; 21 defers
3ph	Remote bus	yes	no	Z2/0.3s	87B primary; 21 back-up
3ph	Local bus	yes	no	Z1/inst	87B primary; 21 Zone1 overlap
ag	Line (bolted)	no	yes	Z1/inst	concurrent
ag	HIF (Z_{arc})	no	yes	none	21 cannot reach HIF
any	Load	no	no	none	All three restrain

- **Heavy load** ($I_{\text{load}} = 2.0$ pu): $Z_m \approx 0.41\angle 25^\circ = 0.37+j0.17$ pu. $\text{Re} = 0.37 < 0.40$: below the blinder threshold; still outside all Mho circles.
- **Test case LOAD_N** ($R = 0.5$ pu): load blinder active ($\text{Re} = 0.5 > 0.40$); no trip confirmed.

The study confirms that for this short line ($|Z_{\text{line}}| = 0.05$ pu), the Mho circle radius (0.045 pu) is small enough that realistic load impedances never enter the zone — the blinder provides a secondary safeguard.

7 Results Summary

12/12 selectivity scenarios correct (100%)

- **Zone boundaries exact**: Zone 1 at $d = 0.800$, Zone 2 at $d = 1.200$, Zone 3 at $d = 1.800$ — all verified to 0.005 resolution.
- **Internal faults (0–79%)**: Zone 1 instantaneous trip, concurrent with 87L.
- **Internal faults (82–100%)**: Zone 2 timed (0.30 s), 87L provides primary instant trip; 21 defers.
- **HIF**: 21 fails (impedance $j2.275$ pu outside Zone 3); 87L is sole detector — demonstrates

complementarity.

- **Load:** all load scenarios correctly blocked (load blinder or outside Mho circle).
- **No false operation:** OoZ, LOAD_N, LOAD_H, AG_HIF all produce no-trip as expected.

8 Conclusions

1. The three-zone Mho relay (21/21P) integrates correctly into the SAMBP protection suite alongside 87B and 87L.
2. **Non-conflicting coordination** is confirmed for all seven fault classes: the primary protection function always acts faster or with higher specificity than the back-up.
3. **Complementary HIF coverage:** 87L detects HIF that 21 cannot reach due to arc impedance; 21 provides robust back-up for bolted and low-resistance faults.
4. **Local bus Zone 1 overlap** is documented and can be resolved with a 21 Zone 1 blocking signal from 87B in a full implementation.
5. **Load blinder** operates correctly for the test network; no load encroachment observed.

8.1 Future Work

1. **TR-19: Simultaneous multi-feeder transfer** — verify 87B/87L/21 coordination during overlapping GOOSE-driven zone reconfigurations.
2. **Adaptive Zone 1 reach for IBR networks** — under high IBR penetration, infeed from inverters modifies the apparent impedance; the Zone 1 reach may need to adapt dynamically.
3. **Quadrilateral characteristic** — replace the Mho circle with a quadrilateral element for better HIF resistive coverage (still coordinated with 87L).
4. **HIL validation** — run all three functions simultaneously on a real-time digital simulator with physical relay hardware to validate the coordination timing.

References

- IEEE std c37.113-2015: Guide for protective relay applications to transmission lines, 2015.
- P. M. Anderson. *Power System Protection*. IEEE Press / Wiley-Interscience, 1999.
- J. L. Blackburn and T. J. Domin. *Protective Relaying: Principles and Applications*. CRC Press, 3rd edition, 2006.
- PhD Candidate. SAMBP TR-15/2026: 87L line differential — voltage-based pi-section correction (mode 1) and threshold recalibration. Technical report, PhD Thesis Supporting Report, 2026a.
- PhD Candidate. SAMBP TR-16/2026: Combined 87B/87L coordination study. Technical report, PhD Thesis Supporting Report, 2026b.
- PhD Candidate. SAMBP TR-17/2026: Adaptive 87L threshold as a function of restraint current i_{rest} . Technical report, PhD Thesis Supporting Report, 2026c.
- S. H. Horowitz and A. G. Phadke. *Power System Relaying*. Wiley, 3rd edition, 2008.
- G. Ziegler. Numerical distance protection: Principles and applications. 2006. Reference for Mho characteristic and zone coordination practice.